

# LUKE K. KIM

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## SUMMARY

Multidisciplinary product development engineer taking consumer electronics from concept through high-volume overseas production across mechanical, electrical, and software. Rapid prototyping, injection-molded parts with ODM partners, custom PCB design and embedded firmware, and root cause analysis. Certified SolidWorks Associate (CSWA).

## EXPERIENCE

**COCCO Hair Professional** | Cleveland, OH  
*Product Development Engineer*

June 2024 – Present

- **Improved blade-gap calibration precision 10x** ( $\pm 0.05$  to  $\pm 0.005$  mm) and **cut cycle time 80%** by building a robotic fixture spanning mechanical design, computer vision, and closed-loop motor control; adopted by the factory as a QC step
- Direct product development across trimmer, clipper, and dryer programs from concept through manufacturing release as first point of contact for technical issues; newest product ships **~20,000 units/month**
- Developing a trimmer/clipper control PCB from the ground up (Altium Designer, ESP32 firmware); built factory proof-of-concepts including a miniaturized analog tachometer clipper (stepper-driven needle, adaptive engine audio, BLDC drive) widely considered infeasible at that size
- Owned a new accessory product end-to-end: consumer research, sourcing, testing, product and packaging design, production launch
- Cut defect rate from **>10% to <5%** on a prior-generation product via root cause analysis of field and line failures; redesigned high-failure switch mechanisms and charging contacts with tooling engineers through injection-mold DFM, cutting return rates
- Led development of a novel low-vibration motor on-site at a supplier's factory in China, achieving **~5x lower vibration** than baseline in a prototype queued for launch; conduct line audits and first-article inspections on repeated visits
- Built standardized test adapters and instrumentation (airflow, acoustic, battery, vibration) to benchmark products against competitors, informing noise-reduction and blade-selection decisions
- Automated internal reporting (1–2 hr/day) and lead an internal team building AI-driven software tools on two-week sprints

**Cambridge Mobile Telematics** | Boston, MA  
*Mechanical Engineering Intern, IoT Devices*

June 2023 – August 2023

- Designed IoT product components, manufacturing fixtures, and 3D-printed test housings in Onshape, improving assembly efficiency and streamlining testing during material changes
- Built a battery charge/discharge test PCB (Altium, Arduino); resolved field and line failures through root cause analysis

**Vanderbilt University — SCALE Program (NASA Grant)** | Nashville, TN  
*Space Applications Research Intern*

December 2020 – May 2024

- Cut testing time **80%** by designing a microscope adapter and custom PCB with LabVIEW automation for radiation-effects research on silicon carbide semiconductors

## PROJECTS

- **Amazon Robotics — Work-in-Progress Buffer (Senior Capstone, 2024):** 4-conveyor buffer with computer vision to organize exception packages for a robotic sorting arm, projected to reduce manual exception handling **75%**
- **Programmable Syringe Pump (2023):** SolidWorks build with custom stepper drive, flow control from **1  $\mu$ L/min to 10 mL/min**

## EDUCATION

**Vanderbilt University** | Nashville, TN  
*Bachelor of Engineering, Mechanical Engineering; Minor in Digital Fabrication*

May 2024

## SKILLS & CERTIFICATIONS

**CAD & Analysis:** SolidWorks (CSWA certified), Onshape, Fusion 360, Ansys, Altium Designer

**Manufacturing:** DFM for additive manufacturing, injection-molded part design (applying DfX feedback), rapid prototyping (3D printing, laser cutting, CNC), fixture design, GD&T, first-article inspection

**Electronics & Software:** Custom PCB design (Altium), Arduino/ESP32 firmware, computer vision, MATLAB, Python, LabVIEW, workflow automation

**Testing & Validation:** Root cause analysis, supplier 8D review, acoustic/vibration/thermal/battery testing, test fixture design

**Languages:** English (native), Spanish (intermediate)